

Relational Database Design Algorithms and Further Dependencies

The two desirable properties of decompositions, namely,

- the **dependency preservation property** and the **lossless (or nonadditive) join property**, which are both used by the design algorithms to achieve desirable decompositions.
- Fourth normal form (4NF) and fifth normal form (5NF)

DESIGNING A SET OF RELATIONS

The Approach of Relational Synthesis (Bottom-up Design):

- Assumes that all possible functional dependencies are known.
- First constructs a minimal set of FDs
- Then applies algorithms that construct a target set of 3NF or BCNF relations.
- Additional criteria may be needed to ensure the the *set of relations* in a relational database are satisfactory

Properties of Relational Decompositions

Relation Decomposition and Insufficiency of Normal Forms:

- Universal Relation Schema:
 - A relation schema $R = \{A_1, A_2, \dots, A_n\}$ that includes all the attributes of the database.
- Universal relation assumption:
 - Every attribute name is unique.

Properties of Relational Decompositions

Relation Decomposition and Insufficiency of Normal Forms (cont.):

- **Decomposition:**
 - The process of decomposing the universal relation schema R into a set of relation schemas $D = \{R_1, R_2, \dots, R_m\}$ that will become the relational database schema by using the functional dependencies.
- **Attribute preservation condition:**
 - Each attribute in R will appear in at least one relation schema R_i in the decomposition so that no attributes are “lost”.

Properties of Relational Decompositions

Another goal of decomposition is to have each individual relation R_i in the decomposition D be in BCNF or 3NF.

Additional properties of decomposition are needed to prevent from generating spurious tuples.

Properties of Relational Decompositions

Dependency Preservation Property of a Decomposition:

- Definition: Given a set of dependencies F on R , the **projection** of F on R_i , denoted by $p_{R_i}(F)$ where R_i is a subset of R , is the set of dependencies $X \rightarrow Y$ in F^+ such that the attributes in $X \cup Y$ are all contained in R_i .
- Hence, the projection of F on each relation schema R_i in the decomposition D is the set of functional dependencies in F^+ , the closure of F , such that all their left- and right-hand-side attributes are in R_i .

Properties of Relational Decompositions

Dependency Preservation Property of a Decomposition (cont.):

- Dependency Preservation Property:
 - A decomposition $D = \{R_1, R_2, \dots, R_m\}$ of R is **dependency-preserving** with respect to F if the union of the projections of F on each R_i in D is equivalent to F ; that is
$$((\pi_{R_1}(F)) \cup \dots \cup (\pi_{R_m}(F)))^+ = F^+$$

Claim 1:

- It is always possible to find a dependency-preserving decomposition D with respect to F such that each relation R_i in D is in 3nf.

Properties of Relational Decompositions

Lossless (Non-additive) Join Property of a Decomposition:

- Definition: Lossless join property: a decomposition $D = \{R_1, R_2, \dots, R_m\}$ of R has the **lossless (nonadditive) join property** with respect to the set of dependencies F on R if, for *every* relation state r of R that satisfies F , the following holds, where $*$ is the natural join of all the relations in D :

$$* (\pi_{R_1}(r), \dots, \pi_{R_m}(r)) = r$$

- Note: The word loss in lossless refers to loss of information, not to loss of tuples. In fact, for “loss of information” a better term is “**addition of spurious information**”

Properties of Relational Decompositions

Lossless (Non-additive) Join Property of a Decomposition (cont.):

Algorithm 11.1: Testing for Lossless Join Property

- **Input:** A universal relation R , a decomposition $D = \{R_1, R_2, \dots, R_m\}$ of R , and a set F of functional dependencies.
- 1.** Create an initial matrix S with one row i for each relation R_i in D , and one column j for each attribute A_j in R .
- 2.** Set $S(i,j) := b_{ij}$ for all matrix entries. (* each b_{ij} is a distinct symbol associated with indices (i,j) *).
- 3.** For each row i representing relation schema R_i
 - {for each column j representing attribute A_j
 - {if (relation R_i includes attribute A_j) then set $S(i,j) := a_j$ };};
 - (* each a_j is a distinct symbol associated with index (j) *)
- CONTINUED on NEXT SLIDE

Properties of Relational Decompositions

4. Repeat the following loop until a complete loop execution results in no changes to S

{for each functional dependency $X \rightarrow Y$ in F

{for all rows in S *which have the same symbols* in the columns corresponding to attributes in X

{make the symbols in each column that correspond to an attribute in Y be the same in all these rows as follows:

If any of the rows has an “a” symbol for the column, set the other rows to that *same* “a” symbol in the column.

If no “a” symbol exists for the attribute in any of the rows, choose one of the “b” symbols that appear in one of the rows for the attribute and set the other rows to that same “b” symbol in the column ;

}; }; };

5. If a row is made up entirely of “a” symbols, then the decomposition has the lossless join property; otherwise it does not.

Properties of Relational Decompositions

Lossless (nonadditive) join test for n -ary decompositions.

(a) Case 1: Decomposition of EMP_PROJ into EMP_PROJ1 and EMP_LOCS fails test.

(b) A decomposition of EMP_PROJ that has the lossless join property.

- (a) $R = \{SSN, ENAME, PNUMBER, PNAME, PLOCATION, HOURS\}$ $D = \{R_1, R_2\}$
 $R_1 = EMP_LOCS = \{ENAME, PLOCATION\}$
 $R_2 = EMP_PROJ1 = \{SSN, PNUMBER, HOURS, PNAME, PLOCATION\}$
 $F = \{SSN \rightarrow ENAME; PNUMBER \rightarrow \{PNAME, PLOCATION\}; \{SSN, PNUMBER\} \rightarrow HOURS\}$

	SSN	ENAME	PNUMBER	PNAME	PLOCATION	HOURS
R_1	b_{11}	a_2	b_{13}	b_{14}	a_5	b_{16}
R_2	a_1	b_{22}	a_3	a_4	a_5	a_6

(no changes to matrix after applying functional dependencies)

(b)

EMP		PROJECT			WORKS_ON		
SSN	ENAME	PNUMBER	PNAME	PLOCATION	SSN	PNUMBER	HOURS

Properties of Relational Decompositions

Lossless (nonadditive) join test for n-ary decompositions.
(c) Case 2: Decomposition of EMP_PROJ into EMP, PROJECT, and WORKS_ON satisfies test.

(c)

$R = \{SSN, ENAME, PNUMBER, PNAME, PLOCATION, HOURS\}$

$D = \{R_1, R_2, R_3\}$

$R_1 = EMP = \{SSN, ENAME\}$

$R_2 = PROJ = \{PNUMBER, PNAME, PLOCATION\}$

$R_3 = WORKS_ON = \{SSN, PNUMBER, HOURS\}$

$F = \{SSN \rightarrow \{ENAME, PNUMBER\}; PNUMBER \rightarrow \{PNAME, PLOCATION\}; \{SSN, PNUMBER\} \rightarrow HOURS\}$

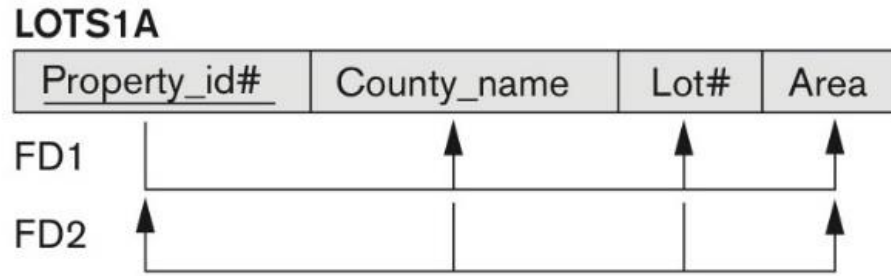
	SSN	ENAME	PNUMBER	PNAME	PLOCATION	HOURS
R_1	a_1	a_2	b_{13}	b_{14}	b_{15}	b_{16}
R_2	b_{21}	b_{22}	a_3	a_4	a_5	b_{26}
R_3	a_1	b_{32}	a_3	b_{34}	b_{35}	a_6

(original matrix S at start of algorithm)

	SSN	ENAME	PNUMBER	PNAME	PLOCATION	HOURS
R_1	a_1	a_2	b_{13}	b_{14}	b_{15}	b_{16}
R_2	b_{21}	b_{22}	a_3	a_4	a_5	b_{26}
R_3	a_1	b_{32} a_2	a_3	b_{34} a_4	b_{35} a_5	a_6

(matrix S after applying the first two functional dependencies - last row is all "a" symbols, so we stop)

(c)



D1(P,L,C,A)

D2(A,P)

D3(C,T)

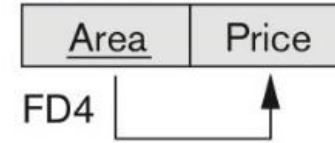
$P \rightarrow LCA$

$A \rightarrow P$

$C \rightarrow T$

$CL \rightarrow AP$

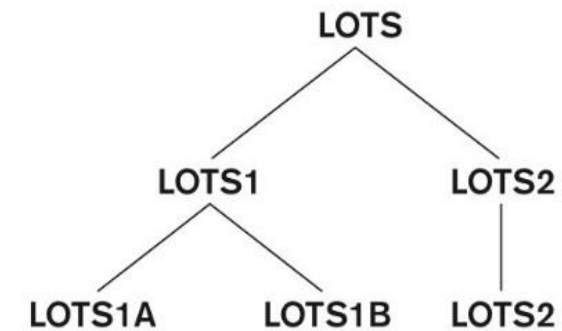
LOTS1B



LOTS2



(d)



1NF

2NF

3NF

Properties of Relational Decompositions

Testing Binary Decompositions for Lossless Join Property

- **Binary Decomposition:** Decomposition of a relation R into two relations.
- **PROPERTY LJ1 (lossless join test for binary decompositions):** A decomposition $D = \{R_1, R_2\}$ of R has the lossless join property with respect to a set of functional dependencies F on R *if and only if* either
 - The f.d. $((R_1 \cap R_2) \rightarrow (R_1 - R_2))$ is in F^+ , or
 - The f.d. $((R_1 \cap R_2) \rightarrow (R_2 - R_1))$ is in F^+ .

Properties of Relational Decompositions

Consider a schema $R(A, B, C, D)$ and functional dependencies $A \rightarrow B$ and $C \rightarrow D$. Then the decomposition of R into $R_1(AB)$ and $R_2(CD)$ is

- A. Dependency preserving and lossless join
- B. Lossless join but not dependency preserving
- C. Dependency preserving but not lossless join
- D. Not dependency preserving and not lossless join

Thank you